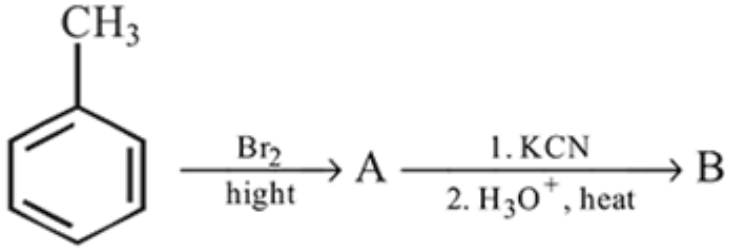
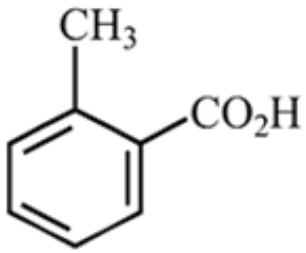
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

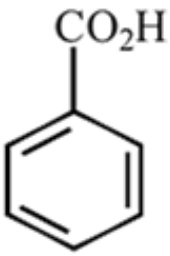
What is the final product (B) of this sequence?



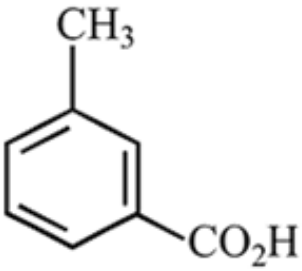
(A)



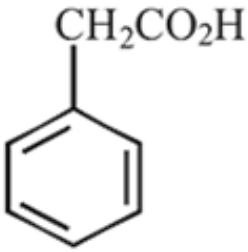
(B)



(C)



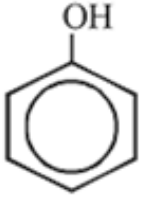
(D)

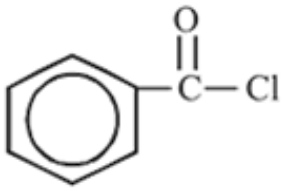


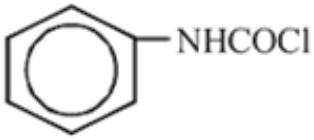
Q.2

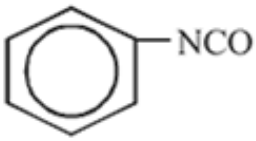
Aniline reacts with phosgene and KOH to form

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(A) 

(B) 

(C) 

(D) 

Q.3

In the reaction $C_6H_5NH_2 + CS_2 \xrightarrow[\Delta]{HgCl_2} \longrightarrow$ the product obtained is

(A) phenyl isocyanide

(B) phenyl cyanide

(C) p -amino benzene

(D) phenyl isothiocyanate sulphonic acid

Q.4

Acetaldoxime reacts with P_2O_5 to give

(A) Methyl cyanide

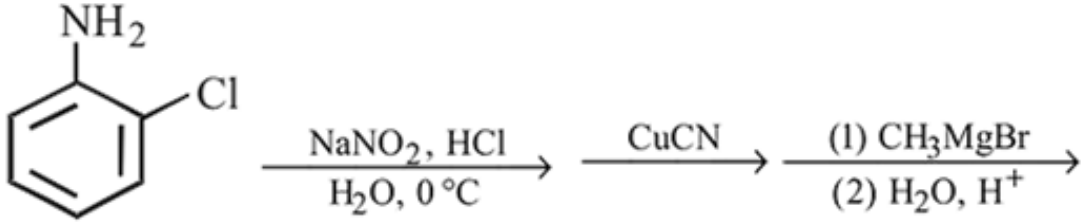
(B) Methyl cyanate

(C) Ethyl cyanide

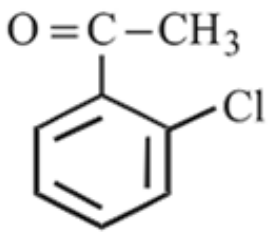
(D) None of these

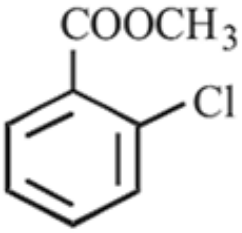
Q.5

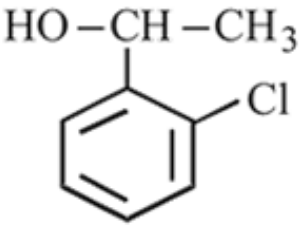
What is the product of the following series of reactions?

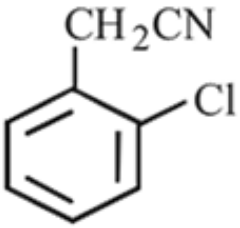


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(A) 

(B) 

(C) 

(D) 

Q.6

Which of the following reactions does not yield an amine

(A) $\mathrm{RX} + \mathrm{NH_3} \rightarrow$

(B) $RCH = NOH + [H] \xrightarrow[C_2H_5OH]{Na}$

(C) $\mathrm{RCN} + \mathrm{H_2O} \xrightarrow{\mathrm{H^+}}$

(D) $RCONH_2 + 4H \xrightarrow{LiAlH_4}$

Q.7

In the mechanism of Hoffmann reaction, which intermediate rearranges to alkyl isocyanate?

(A) Bromamide

(B) Nitrene

(C) Nitroso

(D) Amide

Q.8

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$CH_3CH_2Cl \xrightarrow{NaCN} X \xrightarrow{Ni/H_2} Y \xrightarrow{\text{acetic anhydride}} Z$

Z in the above reacting sequence is

(A) $CH_3CH_2CH_2NHCOCH_3$

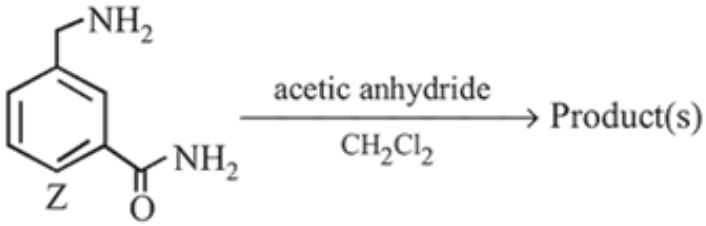
(B) $CH_3CH_2CH_2NH_2$

(C) $CH_3CH_2CH_2CONHCH_3$

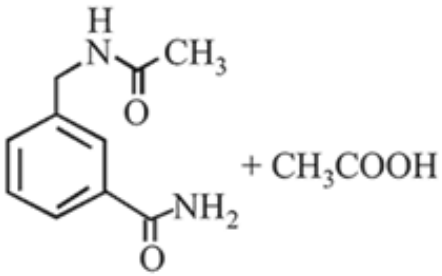
(D) $CH_3CH_2CH_2CONHCOCH_3$

Q.9

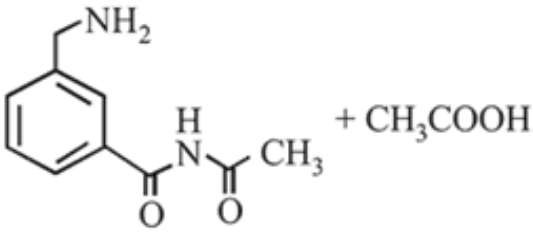
In the reaction shown below, the major product(s) formed is/are



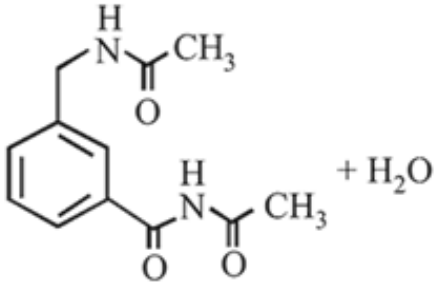
(A)



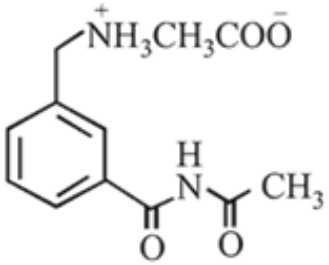
(B)



(C)



(D)



Q.10

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Which of the following statements regarding amides is not correct?

(A) Amides do not form salts when treated with aqueous acids

(B) The aqueous solutions of amides are alkaline

(C) Amides are very poor nucleophiles

(D) Amides are considerably less reactive than acid chlorides

Q.11

When aniline reacts with oil of bitter almonds (C_6H_5CHO) condensation takes place and benzal derivative is formed. This is known as

(A) Million's base

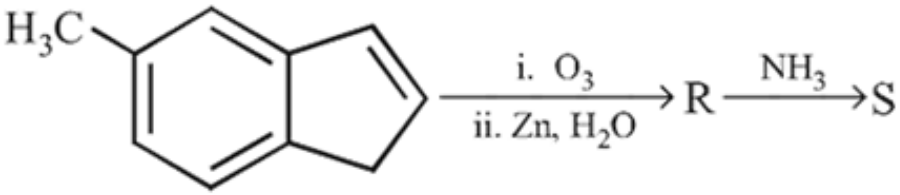
(B) Schiff's reagent

(C) Schiff's base

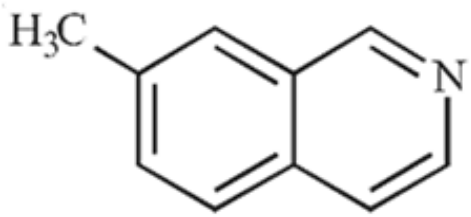
(D) Benedict's reagent

Q.12

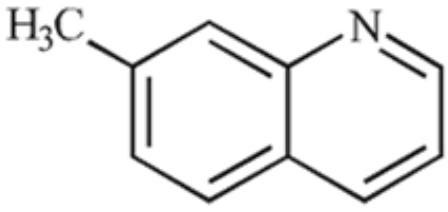
In the following reactions, the product S is



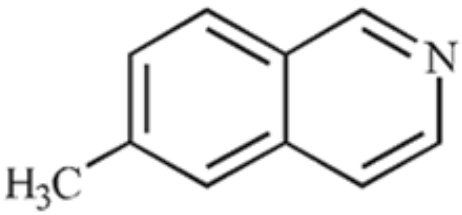
(A)



(B)

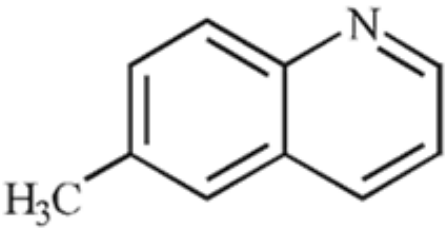


(C)



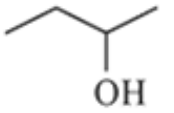
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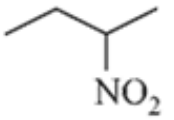
(D)

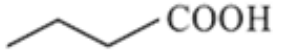


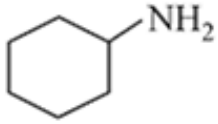
Q.13

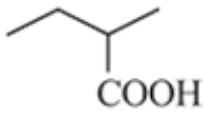
Total number of reactions in which isocyanides are formed

(I)  $\xrightarrow[(2)\ AgCN]{(1)\ PCl_3}$

(II)  $\xrightarrow[(2)\ CHCl_3 / OH^-]{(1)\ Fe + HCl}$

(III)  $\xrightarrow[(2)\ P_2O_5 / \Delta]{(1)\ NH_3 / \Delta}$

(IV)  $\xrightarrow[(2)\ SOCl_2\ (3)\ NaCN]{(1)\ KNO_2 + HCl}$

(V)  $\xrightarrow[(2)\ Br_2 / \Delta\ (3)\ AgCN]{(1)\ AgOH}$

(A) 1

(B) 4

(C) 2

(D) 3

Q.14

Ethyl isocyanide on hydrolysis in acidic medium generates

(A) propanoic acid and ammonium salt

(B) ethanoic acid and ammonium salt

(C) methylamine salt and ethanoic acid

(D) ethylamine salt and methanoic acid

Q.15

An organic compound (A) on reduction gives compound (B). (B) on treatment with $CHCl_3$ and alcoholic KOH gives (C). (C) on catalytic reduction gives N-methylaniline. The compound A is

(A) Methylamine

(B) Nitromethane

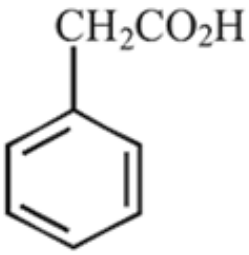
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(C) Aniline

(D) Nitrobenzene

Answers & Solutions

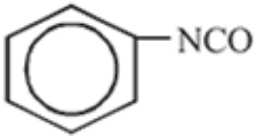
Q.1 Answer:



Solution:

$$(A)\mathrm{Ph}-\mathrm{CH_2}-\mathrm{Br}\xrightarrow{\mathrm{KCN}}\mathrm{Ph}-\mathrm{CH_2}-\mathrm{C}\equiv\mathrm{N}\xrightarrow{\mathrm{H_3O^+}}\mathrm{Ph}-\mathrm{CH_2}-\mathrm{CO_2H}(\mathrm{B})$$

Q.2 Answer:



Solution:

$$\mathrm{C_6H_5NH_2}+\mathrm{COCl_2}+\mathrm{KOH}\longrightarrow$$
$$\mathrm{C_6H_5NH.COCl}+\mathrm{HCl}\xrightarrow{\Delta}\mathrm{C_6H_5NCO}+\mathrm{HCl}$$

Q.3 Answer:

phenyl isothiocyanate sulphonic acid

Solution:

$$\mathrm{C_6H_5NH_2}+\mathrm{CS_2}\rightarrow\mathrm{C_6H_5NH.CS\cdot SH}\xrightarrow{\mathrm{HgCl_2}}$$
$$\mathrm{C_6H_5\ N=C=S}+\mathrm{HgS}+2\mathrm{HCl}$$

The reaction is called mustard oil reaction.

Q.4 Answer:

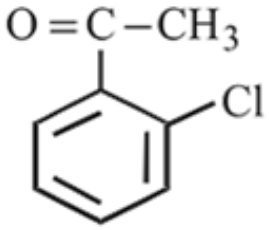
Methyl cyanide

Solution:

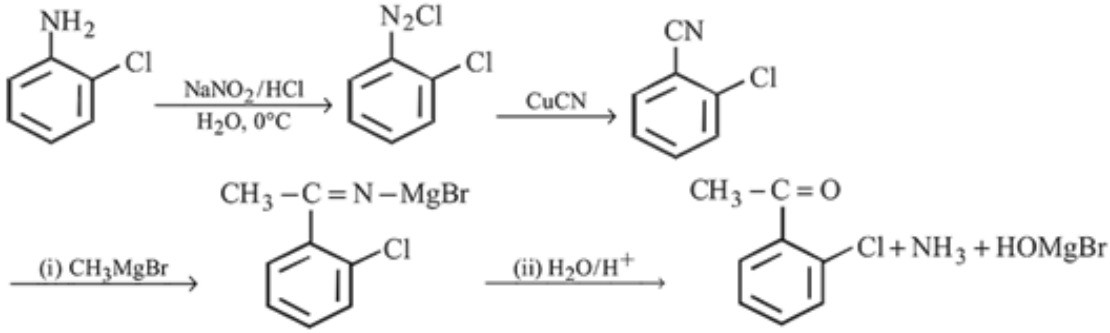
$$\mathrm{CH_3CH=NOH}\xrightarrow{\mathrm{P_2O_5}}\mathrm{CH_3CN}+\mathrm{H_2O}$$

Q.5 Answer:

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Solution:



Q.6 Answer:

$$RCN + H_2O \xrightarrow{H^+}$$

Solution:

$$R - CN + H_2O \xrightarrow{H_2O/H^+} RCOOH + NH_3$$

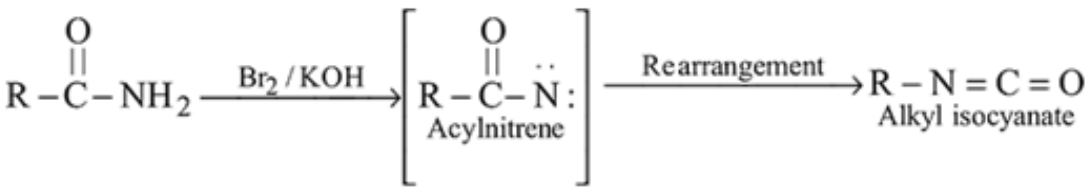
It yield amine when reduced as -

$$R - \mathrm{CN} + \mathrm{H_2} \rightarrow \mathrm{R - CH_2 - NH_2}$$

Q.7 Answer:

Nitrene

Solution:

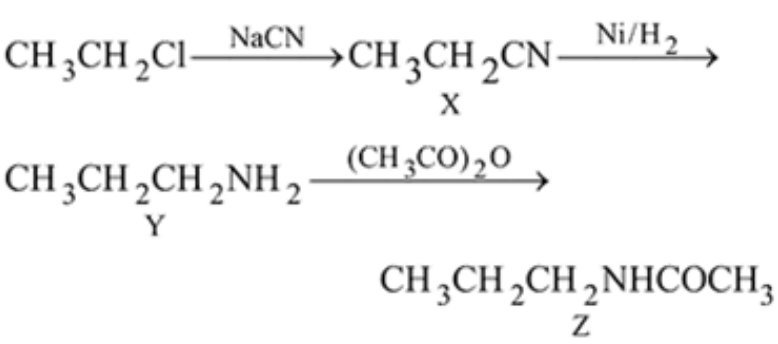


Q.8 Answer:

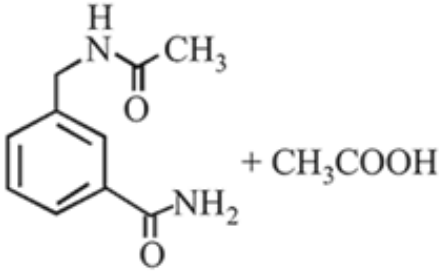
$CH_3CH_2CH_2NHCOCH_3$

Solution:

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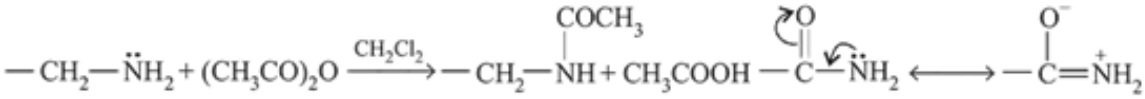


Q.9 Answer:



Solution:

$-\ddot{N}H_2$ group is acetylated by acetic anhydride in methylene chloride (solvent).



Note that $-CONH_2$ group does not undergo acetylation because here lone pair of electrons is delocalised.

Q.10 Answer:

The aqueous solutions of amides are alkaline

Solution:

Amides are not basic and do not form salts in aqueous solution. Amides are quite stable, compared to other acid derivatives, and vigorous conditions are required for their hydrolysis.

Q.11 Answer:

Schiff's base

Solution:

Benzaldehyde reacts with primary aromatic amine to form schiff's base

$$C_6H_5CH=O + C_6H_5NH_2 \longrightarrow$$

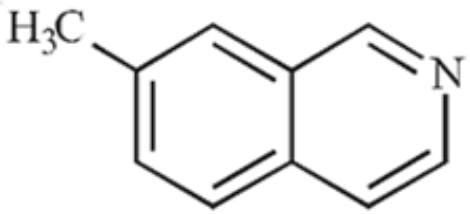
Benzaldehyde Aniline

$$C_6H_5CH=NC_6H_5 + H_2O$$

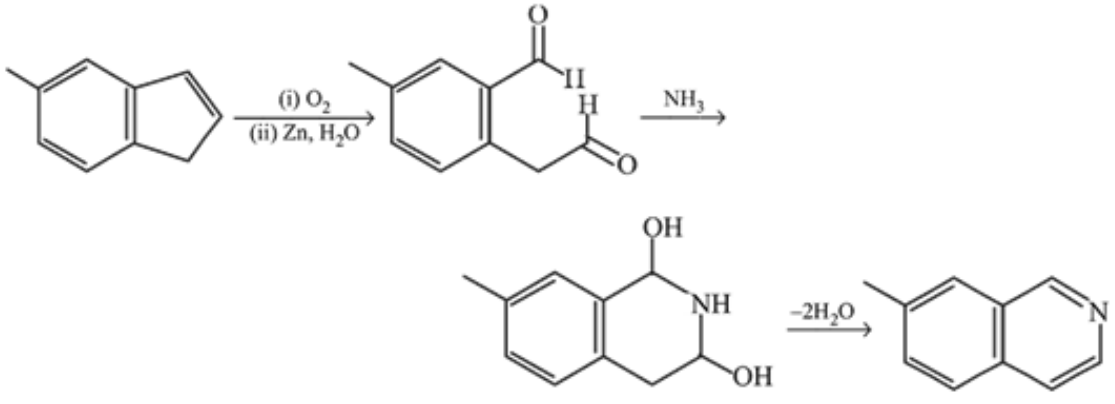
Benzylidene aniline

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Q.12 Answer:



Solution:



Q.13 Answer:

3

Solution:

I, II and V.

Q.14 Answer:

ethylamine salt and methanoic acid

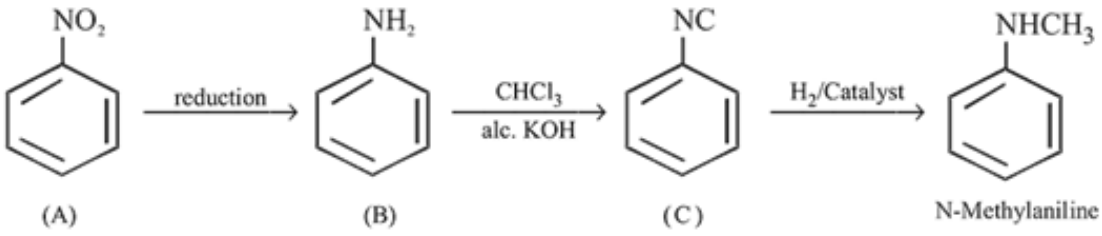
Solution:

$$CH_3CH_2N \equiv C + H_2O \xrightarrow{H^+} CH_3CH_2NH_2 + HCOOH$$

Q.15 Answer:

Nitrobenzene

Solution:



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